

THE HESSIAN ARRANGEMENT REQUIRES FOUR ADDED LINES FOR SUPERSOLVABILITY

ABSTRACT. Let $\mathcal{H}$ be the twelve-line Hessian arrangement in $\mathbb{P}_{\mathbb{C}}^2$. We prove that $\text{extSS}(\mathcal{H}) = 4$. Thus no extension of $\mathcal{H}$ by one line is supersolvable. Since $\mathcal{H}$ is free with exponents $(4, 7)$ and has maximal multiplicity 4, this gives a negative answer to the supersolvable-extension question of Dimca, Kühne, and Pokora. Dimca had already exhibited the relevant exceptional collinearity; the contribution here is the obstruction for every possible modular point and the exact extension number.

1. STATEMENT

Let $\mathcal{A}$ be a line arrangement in $\mathbb{P}_{\mathbb{C}}^2$. A point $p \in \text{Sing}(\mathcal{A})$ is *modular* if the line pq belongs to $\mathcal{A}$ for every $q \in \text{Sing}(\mathcal{A}) \setminus \{p\}$. The arrangement is *supersolvable* if it has a modular point. For line arrangements, that is, in rank three, this is equivalent to supersolvability of the intersection lattice $L(\mathcal{A})$, which is the definition used in [3, Definition 1.1]; see [4], and compare [2], where a line arrangement having a modular point is called supersolvable. Following Kabat [3, Definition 1.3], define

$$\text{extSS}(\mathcal{A}) = \min\{|\mathcal{B} \setminus \mathcal{A}| : \mathcal{A} \subseteq \mathcal{B}, \mathcal{B} \text{ supersolvable}\}.$$

For line arrangements this is equivalent to Kabat's definition by supersolvable resolutions, since the added lines may be ordered arbitrarily.

Let ω be a primitive cube root of unity. The Hessian arrangement is

$$(1) \quad \mathcal{H} : \quad xyz((x^3 + y^3 + z^3)^3 - 27x^3y^3z^3) = 0.$$

Its twelve lines are

$$x = 0, \quad y = 0, \quad z = 0, \quad x + \omega^a y + \omega^b z = 0 \quad (a, b \in \mathbb{Z}/3\mathbb{Z}).$$

Dimca [1, Example 6.5(i)] records that $\mathcal{H}$ is free with first exponent $4 = m(\mathcal{H})$; hence its exponents are $(4, 7)$.

Theorem 1. *The Hessian arrangement satisfies*

$$\text{extSS}(\mathcal{H}) = 4.$$

We recall the question this answers. Let $\mathcal{A}$ be an arrangement of d lines with first exponent d_1 , and let $p \in \mathcal{A}$ be a point of multiplicity m_p with

$$m_p \leq d_1 + 1 < d - m_p + 1.$$

2020 *Mathematics Subject Classification.* 52C35, 32S22.

Key words and phrases. Hessian arrangement, line arrangement, supersolvable arrangement, modular point, extension to supersolvability.

Then [1, Theorem 1.12(2)] states that if $m_p = d_1$ and $m_p = m(\mathcal{A})$, the maximal multiplicity of points of $\mathcal{A}$, then all the other multiple points of $\mathcal{A}$ that are not connected to p by lines of $\mathcal{A}$ lie on a single line L . Assuming $d_1 = m(\mathcal{A})$, Dimca, Kühne, and Pokora ask in [2, Remark 2.3] “whether the line L in [1, Theorem 1.12(2)] can be always chosen such that $\mathcal{A} \cup L$ is supersolvable”. For $\mathcal{H}$ one has $d = 12$ and $d_1 = 4 = m(\mathcal{H})$, and for a point p of multiplicity $m_p = 4$ the hypothesis above reads $4 \leq 5 < 9$; thus $\mathcal{H}$ is in the scope of the question. Theorem 1 gives a negative answer.

2. INCIDENCES

The singular points of $\mathcal{H}$ are

$$\begin{aligned}\mathcal{P} &= \{(0 : 1 : -\omega^j), (1 : 0 : -\omega^j), (1 : -\omega^j : 0) : j \in \mathbb{Z}/3\mathbb{Z}\}, \\ \mathcal{V} &= \{(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1)\} \cup \{(1 : \omega^r : \omega^s) : r, s \in \mathbb{Z}/3\mathbb{Z}\}.\end{aligned}$$

Direct substitution shows that the nine points of $\mathcal{P}$ have multiplicity four, the twelve points of $\mathcal{V}$ have multiplicity two, and each line of $\mathcal{H}$ contains three points of $\mathcal{P}$ and two points of $\mathcal{V}$. Moreover,

$$9 \binom{4}{2} + 12 \binom{2}{2} = 66 = \binom{12}{2},$$

so these are all the singular points.

Lemma 2. *The following hold.*

- (1) *Every two points of $\mathcal{P}$ lie on a line of $\mathcal{H}$.*
- (2) *For each $p \in \mathcal{P}$, exactly four points of $\mathcal{V}$ are not joined to p by a line of $\mathcal{H}$. Those four points are collinear on a line not containing p .*
- (3) *Let $v \in \mathcal{V}$. The two lines of $\mathcal{H}$ through v contain six points of $\mathcal{P}$. If b_1, b_2, b_3 are the remaining points of $\mathcal{P}$, then each line vb_i contains no point of $\mathcal{V}$ other than v .*

Proof. Each line of $\mathcal{H}$ contains three points of $\mathcal{P}$, and

$$12 \binom{3}{2} = \binom{9}{2}.$$

No pair is counted twice, since two distinct points determine a unique projective line. This proves (1).

Coordinate permutations and diagonal maps

$$(x : y : z) \longmapsto (\omega^a x : \omega^b y : \omega^c z)$$

preserve $\mathcal{H}$ and act transitively on $\mathcal{P}$. For (2), it therefore suffices to take $p = (1 : -1 : 0)$. The four exceptional points are

$$(0 : 0 : 1), \quad (1 : 1 : 1), \quad (1 : 1 : \omega), \quad (1 : 1 : \omega^2).$$

They lie on $x - y = 0$, which does not contain p .

The same symmetries have two orbits on $\mathcal{V}$: the coordinate vertices and the points with all coordinates nonzero. For $v = (0 : 0 : 1)$, the two old lines through v are $x = 0$ and $y = 0$, and the remaining points of $\mathcal{P}$ are

$$b_j = (1 : -\omega^j : 0), \quad j \in \mathbb{Z}/3\mathbb{Z}.$$

Their joins with v are $\omega^j x + y = 0$. None contains another point of $\mathcal{V}$: substituting $(1 : 0 : 0)$ gives $\omega^j \neq 0$ and substituting $(0 : 1 : 0)$ gives $1 \neq 0$, while at $(1 : \omega^r : \omega^s)$ the equation would force $\omega^j + \omega^r = 0$, which is impossible for cube roots of unity.

For $v = (1 : 1 : 1)$, the remaining points of $\mathcal{P}$ are

$$(0 : 1 : -1), \quad (1 : 0 : -1), \quad (1 : -1 : 0),$$

and their joins with v are

$$2x - y - z = 0, \quad x - 2y + z = 0, \quad x + y - 2z = 0.$$

The first contains no coordinate vertex. At a point $(1 : \omega^r : \omega^s)$, its equation gives $\omega^r + \omega^s = 2$, hence $r = s = 0$. The other two cases follow by permuting the coordinates. This proves (3). $\square$

3. PROOF

Let $\mathcal{B} \supseteq \mathcal{H}$ be supersolvable, set $k = |\mathcal{B} \setminus \mathcal{H}|$, and let X be a modular point of $\mathcal{B}$.

Suppose first that $X \in \mathcal{P}$. By Lemma 2(2), there are four points $v_1, \dots, v_4 \in \mathcal{V}$ not joined to X by old lines. Modularity forces the four joins Xv_i to belong to $\mathcal{B}$. They are outside $\mathcal{H}$ and are distinct: the points v_i lie on a line not containing X , so no line through X contains two of them. Thus $k \geq 4$.

Suppose next that $X \in \mathcal{V}$. The two old lines through X contain six points of $\mathcal{P}$. The remaining three points require three distinct added lines through X , because a line outside $\mathcal{H}$ contains at most one point of $\mathcal{P}$ by Lemma 2(1). By Lemma 2(3), none of these three lines contains a point of $\mathcal{V} \setminus \{X\}$. The union of the two old lines through X contains only three points of $\mathcal{V}$, including X . Hence some point of $\mathcal{V}$ requires a fourth added line through X , and again $k \geq 4$.

Finally, suppose that $X \notin \text{Sing}(\mathcal{H})$. Every point of $\mathcal{P}$ remains singular in $\mathcal{B}$, so modularity requires its join with X to belong to $\mathcal{B}$. At most one old line passes through X , and it contains at most three points of $\mathcal{P}$. Every added line through X contains at most one point of $\mathcal{P}$. If $k \leq 3$, at most six of the nine points of $\mathcal{P}$ can be joined to X , a contradiction. Therefore

$$\text{extSS}(\mathcal{H}) \geq 4.$$

The three cases give lower bounds only; the individual counts are not claimed to be sharp, and $k \geq 4$ is all that is needed.

For the reverse inequality, take $p = (1 : -1 : 0)$ and set

$$\mathcal{B} = \mathcal{H} \cup \{x + y = 0\} \cup \{x + y - 2\omega^j z = 0 : j \in \mathbb{Z}/3\mathbb{Z}\}.$$

These are precisely the four joins from p to the exceptional points in Lemma 2(2). Every old singular point is joined to p by a line of $\mathcal{B}$. If $q \in \text{Sing}(\mathcal{B}) \setminus \text{Sing}(\mathcal{H})$, then the intersection defining q involves an added line; hence q lies on an added line through p . Thus p is modular in $\mathcal{B}$, and

$$\text{extSS}(\mathcal{H}) \leq 4.$$

This proves Theorem 1.

For $p = (1 : -1 : 0)$, the four singular points not joined to p by lines of $\mathcal{H}$ are distinct and determine the unique line $x - y = 0$. Thus this is the line supplied by [1, Theorem 1.12(2)]. Since no one-line extension of $\mathcal{H}$ is supersolvable, this proves the claimed negative answer to [2, Remark 2.3].

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